

SENIOR ARCHITECTURE RECOMMENDATION

Required Tools and Technology Guide

A low-maintenance, Azure-first stack for learning AI/NLP and shipping OLGA Connect

Audience: Founder, architect and founding engineering team

Horizon: Learning build -> controlled pilot -> event launch

Cost posture: Use free tiers for development; pay only for event-grade capacity

Status: Recommended baseline - verify region and price before purchase

Primary recommendation: Build a modular monolith on Azure Container Apps with .NET 10, React Native with Expo development builds, Azure SQL for transactions and vectors, and Azure OpenAI only for embeddings. Do not begin with Kubernetes, microservices, Azure AI Search, Redis, an LLM agent or custom phone authentication.

1. Decisions made from the supplied blueprints

Area	Decision	Reason
Launch	Malaysia-first; one region near Kuala Lumpur	The named launch vehicle is in Kuala Lumpur. One saturated market reduces operational ambiguity.
Context join	QR/code first; geofence optional	QR works indoors and avoids background-location complexity.
Authentication	Email OTP with Entra External ID	Lowest-maintenance Azure path. Native customer auth supports email OTP/password, not phone-only primary sign-in.
Matching	Rules plus embeddings	No training data exists; outcomes should create labels first.
Database	Azure SQL including vectors	One managed service handles transactions, geography and embeddings.
Realtime	Polling for learning; Web PubSub at event scale	Avoid fixed realtime capacity until concurrent usage requires it.
File sharing	Deferred from v1	The flow brief defers it; this removes malware and retention risk.
Offline AI	Precomputed embeddings; on-device scoring	No on-device language model is needed.

2. Recommended stack at a glance

Need	Use now	Azure / tool	Cost posture
Mobile	React Native + Expo dev builds + TypeScript	Expo Router, SQLite, SecureStore, Notifications, Location	Open source; app-store accounts separate.

Need	Use now	Azure / tool	Cost posture
Admin	React + Vite	Static Web Apps Free	Good for internal MVP; no free-tier SLA.
API	ASP.NET Core 10 Minimal APIs	Container Apps Consumption	Scale-to-zero and monthly free grant.
Worker	In-process first; split when needed	Queue Storage + Container Apps Job	Very low usage cost; event-driven scale.
Data	EF Core CRUD; SqlClient/Dapper vectors	Azure SQL free offer / GP serverless	100,000 vCore-sec + 32 GB data and backup monthly; no SLA.
Semantic NLP	text-embedding-3-small	Azure OpenAI / Microsoft Foundry	Pay per input token; short intents keep usage small.
Vector retrieval	VECTOR(1536) + cosine distance	Azure SQL	No separate vector database or AI Search.
Identity	Email OTP	Entra External ID	Basic MAU free allowance; SMS charged separately.
Push	APNs + FCM abstraction	Notification Hubs	Free: 500 devices / 1M pushes; Basic for launch.
Realtime chat	Managed WebSocket service	Web PubSub	Free is 20 connections; Standard for hundreds.
Storage	Photos, exports, future files	Blob Storage	Consumption based; lifecycle deletion.
Secrets	Managed identity	Key Vault Standard	Low transaction cost; cache resolved secrets.
Moderation	Rules + classifier	Content Safety F0	Small free allowance; throttles at limit.
Telemetry	OpenTelemetry	Application Insights	Sampling and a conservative daily cap.
Delivery	GitHub Actions + Bicep + azd	Azure Resource Manager	Use workload identity, not saved cloud secrets.

3. Local development tools

Category	Recommended tools	Purpose
Editor	VS Code: C#, Python, ESLint, Bicep	One lightweight environment for mobile, API, AI labs and infrastructure.
Mobile	Node.js LTS, pnpm, Expo CLI, Android Studio, Xcode	Build iOS/Android from one codebase. Use development builds for native features.
Backend	.NET 10 SDK, Docker Desktop, SQL extension	API, worker, migrations and local SQL tests.
AI/NLP	Python 3.12, uv, JupyterLab, pandas, numpy, scikit-learn, sentence-transformers	Learn normalization, embeddings, cosine similarity, ranking and metrics locally.
API	Bruno/Postman, OpenAPI	Exercise endpoints and version request examples.
Testing	xUnit, Testcontainers, Playwright, Maestro, k6	Unit, integration, admin UI, mobile flow and load tests.

Category	Recommended tools	Purpose
Delivery	Git, GitHub Issues/Projects	Code review, backlog, ADRs and release tags.

4. AI and NLP learning toolkit

Goal	Hands-on tool	Product exercise	Evidence
Text basics	Python, regex, simple tokenization	Clean 140-character intents; language and PII checks.	Normalization tests and error examples.
Embeddings	sentence-transformers locally; Azure OpenAI in cloud	Compare offers, needs and paraphrases.	Similarity notebook with labels.
Retrieval	numpy/scikit-learn, then Azure SQL	Reciprocal candidates inside one context.	Precision@5 and Recall@20.
Ranking	Python rules, then .NET	Category complementarity + two semantic directions + freshness.	Versioned weights and golden tests.
Explainability	Reason codes + templates	One reason sentence for each card.	100 percent explanation coverage.
Evaluation	pandas + scikit-learn metrics	At least 200 labels split by member/org.	Gold set, error taxonomy and model card.
Responsible AI	Content Safety + PII minimization	Prevent private chat/location from reaching models.	Threat tests and data-flow review.

5. Azure SQL and vector recommendation

Use Azure SQL instead of PostgreSQL: Store business data and embeddings in one Azure SQL Database. Use VECTOR(1536) for text-embedding-3-small and exact VECTOR_DISTANCE cosine search after filtering to the active context. This is transactional, inexpensive and sufficient for event-scale candidate sets.

- Use EF Core migrations for relational schema; Microsoft.Data.SqlClient 6.1+ or Dapper for vector queries.
- Keep float32 vectors. Do not depend on float16 or approximate VECTOR_SEARCH while preview-gated.
- Index context_id, presence expiry, intent status, block pairs and connection status so the vector scan stays small.
- Version the embedding model, dimensions, normalized hash and ranking configuration on every match.

6. Authentication decision - important product change

Do not build phone-only OTP yourself: Entra External ID native customer authentication supports email OTP and email/password. SMS is a charged verification factor, not a drop-in phone-only primary identity. Use email OTP for the MVP. If phone-only is non-negotiable, run a managed-CIAM vendor/security evaluation.

Option	Maintenance	Cost	Recommendation
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Option	Maintenance	Cost	Recommendation
Entra email OTP	Low	Free allowance at small MAU	Use for MVP.
Entra + SMS MFA	Low	Per SMS authentication event	Use selectively, not every sign-in.
Custom OTP + Communication Services	High security responsibility	SMS + engineering/abuse cost	Do not use as identity system.
Managed CIAM with passwordless SMS	Medium vendor dependency	Per MAU/SMS	Evaluate only if phone-only is fixed.

7. Defer or avoid initially

Do not buy/build yet	Why	Adopt when
AKS / Kubernetes	Cluster and networking overhead.	Many independently owned services or hard platform needs.
Microservices	Distributed transactions and deployments slow a small team.	Clear team or independent scaling boundaries.
Azure AI Search	Duplicate vector storage and fixed service cost.	Millions of records or measured ANN/hybrid search need.
Azure Managed Redis	Another dependency; SQL/local cache are enough.	Measured hot-read or coordination bottleneck.
Service Bus Premium	Queue Storage handles simple jobs cheaply.	Ordering, topics, transactions or duplicate detection required.
API Management	Fixed cost/configuration not needed for MVP.	External API products, partners or central policy.
Generative agent	Not required for matching; larger safety surface.	Validated user job after core loop works.
Fine-tuning	No dataset exists.	Thousands of reviewed labels show a stable failure.
File sharing	Malware, moderation and retention obligations.	Demand and policy justify it.
Precise location history	Privacy, battery and trust risk.	Never for this product; use bounded presence.

8. Cost-conscious subscription plan

Directional planning envelopes, not quotes. They exclude app-store accounts, domain, taxes, SMS and staff time. Prices vary by Azure region and agreement.

Stage	Configuration	Monthly envelope	Guardrails
Learning	Free SQL, scale-to-zero API, Static Web Apps Free, Content Safety F0, email OTP	USD 0-25	One dev environment; delete experiments; AI budgets.
Small pilot	Serverless SQL overage, low API traffic, <500 push devices	USD 25-100	Sample logs; no Web PubSub unless needed.
Event launch	Warm API, Web PubSub Standard, Notification Hubs Basic, raised SQL capacity	USD 100-400 + SMS	Scale up just before; load test; scale down after.

Stage	Configuration	Monthly envelope	Guardrails
Growth	Multiple replicas, paid realtime/push and stronger DR	Measure first	Track cost per active user and event.

- Set subscription budgets at 50/80/100 percent and service alerts for SQL allowance, telemetry and AI usage.
- Use minReplicas=0 for dev; temporarily use 1 during the launch window.
- Start without VNet/private endpoints unless security review requires them.
- Tag resources with product, environment, owner, costCenter, dataClass and expiryDate.

9. Suggested repository structure

Area	Contents
apps/mobile	Expo app, offline SQLite schema, outbox and device integrations.
apps/admin	React/Vite staff console and live-ops dashboard.
services/api	Modular monolith: profiles, contexts, intents, matches, connections, chat and moderation.
services/worker	Embedding, match generation, expiry, notifications and outcomes.
packages/contracts	OpenAPI TypeScript client, schemas and reason-code catalog.
ai/notebooks	Anonymized data, baseline, evaluation and error analysis.
infra	Bicep modules, environment parameters, budgets and dashboards.
docs/adr	Identity, offline, retention and ranking decisions.

10. Twelve-week learn-and-build sequence

Weeks	Learn	Build	Exit check
1-2	Python, normalization, cosine, privacy	Anonymized intents + 200 labels	Repeatable baseline notebook.
3-4	React Native, .NET API, SQL	OTP, profile, context, intent, expiry	iOS/Android onboarding.
5-6	Embeddings, metrics, rules	SQL vectors, reciprocal score, reasons	Precision@5 and reason coverage.
7-8	Offline sync, idempotency, push	Context cache, outbox, requests	Airplane-mode script passes.
9-10	WebSockets, moderation, telemetry	Chat, block/report, admin, outcomes	Consent/access tests pass.
11-12	Load, threat, cost testing	Seeded users, dashboards, rollback	Event dress rehearsal.

11. Accounts and subscriptions checklist

- Azure subscription with budgets and least-privilege roles.
- Entra external tenant for members; workforce tenant for administrators.
- Azure OpenAI access and embedding model availability in the selected region.

- Apple Developer and Google Play Console accounts; APNs and FCM credentials.
- GitHub organization with branch protection, secret scanning and OpenID Connect to Azure.
- Verified sender domain; SMS registration only if later required.
- Qualified privacy counsel for Malaysia consent, retention and deletion evidence.

Reference sources

Reviewed 27 August 2026. Azure capabilities, quotas and prices vary by region and agreement; confirm in the Azure pricing calculator before purchase.

[Azure SQL Database free offer FAQ](#)

[Azure SQL vector functions](#)

[Azure SQL vector data type](#)

[Azure OpenAI embeddings tutorial](#)

[Azure Container Apps billing](#)

[Azure Container Apps scaling](#)

[Microsoft Entra External ID authentication methods](#)

[Microsoft Entra External ID pricing concepts](#)

[Azure Web PubSub pricing and capacity](#)

[Azure Notification Hubs FAQ](#)

[Azure Static Web Apps plans](#)

[Azure AI Content Safety overview](#)

[Azure Monitor daily cap](#)

[.NET support policy](#)

[React Native environment guidance](#)